

SQL practices

SQL Question and Answers

1. How to retrieve the second highest salary of an employee?

```
select max(salary) from employee
where salary < (select max(salary) from employee);
```

```
select salary from (select salary, row_number() over(order by salary desc)
as rn
from employee ) as ranked
where rn =2
```

```
select salary from (select salary, dense_rank() over(order by salary desc) as
s rn
from employee) as ranked
where rn = 2
```

2. how to get the nth highest salary in ?

```
select salary from (select salary, dense_rank(order by salary desc) as rn
from employee) as ranked
where rn = N;
```

3. How do you fetch all employees whose salary is greater than the average salary?

```
select * from employee
where salary > (select avg(salary) from employee);
```

4. write a query to display the current date and time in

```
select current_timestamp;
```

5. How to find duplicate records in a table ?

```
select empid, count(*) as count_emp from employee
group by empid
having count(*) > 1
```

6. how can delete duplicate row in ?

```
with cte as(
select id, row_number() over(partition by email order by id) as rn
from employee)

delete from employee
where id in (select id from cte where rn > 1)
```

7. How to get the common records from two tables?

```
select * from employee e
inner join manager m
on m.emp_id = e.emp_id
```

```
select * from employee
where id in (select id from manager)
```

8. How to retrieve the last 10 records from a table?

```
select * from employee order by emp_id desc limit 10
```

9. How do you fetch the top 5 employees with the highest salaries?

```
select * from employee order by salary desc limit 5;
```

10. How to calculate the total salary of all employee?

```
select sum(salary) from employee;
```

11. How to write a query to find all employees who joined in the year 2020?

```
select * from employee where year(date_joining) = 2020
```

```
select * from employee where extract(year from date_joining) = 2020
```

```
select * from employee where date_joining between '2020-01-01' and '2020-12-31'
```

12. write a query to find employees whose name starts with 'A'

```
select * from employee where name like 'A%'
```

13. How can you find the employees who do not have a manager?

```
select e.* from employee e  
join manager m  
on m.mid = e.mid  
where m.mid is null
```

14. How to find the department with the highest number of employees ?

```
select d.name, count(*) as count_emp from employee as e  
join department d  
on e.id = d.id  
group by count_emp  
order by id desc  
limit 1;
```

15. How to get the count of employees in each department?

```
select d.name as dep_name, count(*) as count_emp from employee as e  
left join department as d  
on e.id = d.id  
group by d.name
```

16. write a query to fetch employees having the highest salary in each department?

```

SELECT e.*, d.name AS department_name
FROM employee e
JOIN department d ON e.department_id = d.id
WHERE e.salary = (
    SELECT MAX(salary)
    FROM employee
    WHERE department_id = e.department_id
);

```

```

SELECT *
FROM (
    SELECT e.*, d.name AS department_name,
           ROW_NUMBER() OVER (PARTITION BY e.department_id ORDER BY e.
salary DESC) AS rn
    FROM employee e
    JOIN department d ON e.department_id = d.id
) t
WHERE rn = 1;

```

17. How can you find employees whose salary is between 50,000 and 1,00,000

```

SELECT *
FROM employee
WHERE salary BETWEEN 50000 AND 100000;

```

18. How to find the youngest employee in the organization

```

select * from employee where dob = (select max(dob) from employee)

```

19. How to fetch the first and last records from a table?

```

(select * from employees order by employee_id asc limit 1)
union all
(select * from employees order by employee_id desc limit 1);

```

20. write a query to find all employee who report to a specific manager.

```
select * from employee where manager id = ""
```

21. How can you find the total number of departments in the company?

```
select count(*) AS total_departments  
from department;
```

22. How to find the department with the lowest average salary?

```
select d.name as dep_name, avg(e.salary) as avg_salary from employee as  
e  
inner join department d  
on e.emp_id = d.emp_id  
group by d.name  
order by avg_salary
```

23. How to delete all employee from a department in one query?

```
delete form department where dep_id = 10;
```

24. How to display all employees who have been in the company for more than 5 years?

```
select * from employees where data_of_joining <= current_date - interval 5  
year
```

```
select * from employees where date_diff('year', date_of_joining ,current_da  
te) >5
```

25. How to write a query to remove all records from a table but keep the table structure?

```
truncate from employee
```

26. How to get the current month's name from ?

```
SELECT format_datetime(current_date, 'MMMM') AS month_name;
```

28. How to convert a string to lowercase in ?

```
select lower(name) as emp_name from employee
```

29. write a query to calculate the total sales per customer in a sales table.

```
select cust_id, sum(sales) as total_sales from sales  
group by cust_id
```

30. How to find the second highest salary for each department?

```
select id, salary from (  
  select id, row_number() over(partition by id order by salary desc) as rnk  
  from employee) as ranked  
where rnk = 2
```

31. write a query to fetch all employees whose name end with 'n'.

```
select * from employees where name like '%n'
```

32.